Lounes Benali

Deep Learning Engineer | Speech & Multimodal AI | Large-Scale Training & Inference

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Profile

Speech and multimodal AI engineer working across large-scale data curation, distributed GPU training, model adaptation, rigorous evaluation, and low-latency inference. At MBZUAI's Institute of Foundation Models, co-developing multilingual ASR for dialectal and code-switched speech on Slurm-managed AMD MI210 and NVIDIA H200 clusters.

Professional Experience

Speech AI Research Intern — Feb 2026–Aug 2026 (expected)
Institute of Foundation Models (IFM), MBZUAI — *Paris, France*
Open foundation-model research lab at MBZUAI.

- Core contributor in a four-person speech team co-developing multilingual ASR for dialectal and code-switched speech. The first checkpoint ranked first in CER and third in WER on the public Open Universal ASR Leaderboard—within 2 WER points of leaders Cohere and Audar in Arabic—while supporting diacritized transcription, dialect identification, and code-switching across French, English, Spanish, and other languages.
- Engineered a resilient audio-ingestion system that collected nearly 4M hours, using deduplicated leased SQLite queues, partitioned VPN pools, pipelined download/Opus encoding, tmpfs-first I/O with disk fallback, and remotely verified dataset-shard uploads.
- Built a multi-model curation pipeline over 220M speech segments (roughly 1M hours), combining dialect ID, acoustic-quality labels, ASR hypotheses, forced alignment, and token confidence to produce a compact, high-quality 17k-hour ASR corpus.
- Built and operated large-scale Slurm workflows for distributed training and fault-tolerant inference across AMD/ROCm and NVIDIA/CUDA GPU clusters.
- Engineered a WebSocket streaming gateway around a non-streaming ASR model, combining causal VAD, language-aware hypothesis stabilization, final-first scheduling, and failure-aware endpoint leases; a 16-session replay achieved 1.33 s p95 first-partial and 0.55 s p95 post-EOS latency.
- Ran iterative self-supervised VietASR pretraining from random initialization plus supervised studies using as little as 100 labeled hours; scaled selected runs across 8–16 Slurm nodes and achieved 33.22 WER / 15.10 CER on a balanced 10-hour, 19-dialect evaluation set.
- Reimplemented a pinned FunASR FSMN-VAD frontend and causal encoder as batched PyTorch/ROCm operations, reaching 9,668× real-time end-to-end throughput on a single 8× AMD MI210 node after initialization.
- Ongoing work focuses on developing streaming-oriented audio encoders from scratch to support the broader IFM speech stack.

Deep Learning Engineer — Sep 2024–Feb 2026
Ampere Software Technology (Renault Group) — Master's Work-Study Program — *Guyancourt, France*

- Developed and fine-tuned TTS and speech-to-speech systems (Moshi), contributing from dataset generation and processing through model validation and deployment.
- Built an internal real-time inference platform serving multiple fine-tuned models, voices, versions, and languages; implemented streaming, model compression, inference acceleration, and latency instrumentation.
- Built a synthetic-conversation data pipeline that orchestrated two real-time voice agents, routed audio between them, converted sample rates, and recorded speaker-separated stereo WAVs for speech-model training.
- Built Kotlin/Android Automotive voice-assistant prototypes with coroutine-based WebSocket clients, full-duplex PCM audio, interruption control, jitter-aware playback, and pre-warmed session management.

Technical Staff – Paris 2024 Olympic Games — May 2024–Aug 2024
International Broadcast Center (IBC) — *Le Bourget, France*

- Deployed and supported computers, peripherals, and networked devices on the IBC LAN; diagnosed endpoint and connectivity issues during daily broadcast operations.

Electrical Engineering Intern — Jun 2023–Aug 2023
Schneider Electric — *Algiers, Algeria*

- Studied industrial variable-frequency-drive architectures and evaluated opportunities for machine-learning-based motor-control optimization.

Selected Technical Work

vLLM Audio-Model Integrations on ROCm – IFM-Paris Python, PyTorch, vLLM, ROCm, Slurm

- Implemented an out-of-tree HiggsAudio3Model plugin for vLLM, reproducing its chunked Whisper frontend and stride-2 projector, injecting audio embeddings into the native Qwen3 decoder, and exposing OpenAI-compatible transcription.
- Replaced dynamically loaded remote configuration with importable classes compatible with vLLM EngineCore process serialization; validated greedy token/text parity against Transformers and deployed eight BF16 MI210 endpoints with Slurm lifecycle tooling.
- Separately source-built vLLM for ROCm/gfx90a, integrated Audex/Nemotron-H, and brought up BF16 tensor-parallel audio serving across eight MI210 GPUs.

Seenitt – Streaming Visual AI for Smart Glasses – MentraOS Grant Python, TensorRT, CUDA, GitHub

- Received a MentraOS developer grant for Seenitt; built an Android-to-Python visual-AI stream that H.264-encodes camera frames on-device, decodes them with PyAV over WebSockets, and returns structured results.
- Implemented a streaming chess-assistant prototype using SAM 3 to segment the physical board and pieces, reconstruct and track its 9×9 grid, stabilize observations against legal moves, emit FEN state, and return Stockfish suggestions.
- Optimized the FP16 TensorRT path with cached text embeddings, reused vision features, page-locked buffers, and asynchronous CUDA transfers; overlapped GPU piece decoding with CPU grid reconstruction and tracked all 81 intersections using Lucas–Kanade optical flow.
- Developing a vegetable-recognition prototype using prompt-driven segmentation, ViT classification, and best-view cataloguing, alongside a Blackjack companion prototype.

k2 ROCm Port for Speech Training – IFM-Paris C++, HIP, PyTorch, ROCm, GitHub

- Ported upstream k2's CUDA-oriented C++/PyTorch extension to ROCm/HIP on AMD MI210, preserving the upstream API while adapting its build, packaging, and runtime dispatch.
- Integrated PyTorch device/stream semantics, HIP/hipCUB compatibility, and kernel fixes for ragged-label prefix scans; validated representative FSA and autograd operations.

Android Automotive Voice-AI Prototypes – Ampere Kotlin, Coroutines, WebSockets, Hume GitHub

- Built Kotlin Android Automotive clients for hosted conversational-AI services, including Hume EVI, with 16-kHz PCM capture, buffered 24/48-kHz playback, live transcripts and prosody, interruption control, and tool-call flows.
- Designed concurrency-safe pre-warmed session pools with atomic capacity control, readiness tracking, and language isolation; implemented RMS voice detection, jitter-buffer recovery, and phone, wired, Bluetooth, and automotive audio routing.

Education

Master's Degree in Intelligent Systems Engineering Sep 2024–Present
Sorbonne University *Paris, France*

Bachelor's Degree in Electrical Engineering and Computer Science Sep 2023–May 2024
Paris-Saclay University *Paris, France*

- Graduated top of the class.

Engineering Cycle – Electronics Engineering and Computer Science Sep 2022–Jun 2023
National Polytechnic School of Algiers *Algiers, Algeria*

Preparatory Classes in Science and Technology Sep 2020–Jun 2022
Two-year intensive national competitive-exam program *Algeria*

- Ranked 29th out of 1,576 candidates in the national competitive entrance examination for engineering schools.

Technical Skills

Speech & Multimodal AI: ASR, TTS, speech-to-speech, speech SSL, PyTorch, k2, Hugging Face
Distributed ML & Inference: Slurm, DDP, FSDP, vLLM, AMD/ROCm, NVIDIA/CUDA, multi-node training and inference
Vision & Edge Inference: SAM 3, TensorRT, ONNX Runtime, OpenCV, H.264/PyAV, optical flow
Programming: Python, C, C++, Kotlin, MATLAB, VHDL
Data & Systems: PostgreSQL, SQLite, DuckDB, Parquet, WebDataset, Docker, Git, WebSockets
Hardware & Embedded: FPGA/VHDL, Autodesk EAGLE PCB design
Technical Familiarity: *Audio encoding & representations* — Whisper-style and Qwen AuT encoders, Conformer/FastConformer, streaming causal encoders, log-Mel/fbank, continuous, RVQ/Mimi-tokenized, and hybrid audio representations, audio-to-LLM adapters; *Modeling & objectives* — MoE, cross-modal attention, CTC, RNN-T, AED, diffusion, flow matching (FM, OT-CFM, radial-angular transport); *Adaptation & post-training* — LoRA, PPO/GRPO-style RLHF, DPO; *Efficient training & inference* — model parallelism, FlashAttention, sliding-window attention, KV caching, PagedAttention, speculative decoding, continuous batching, pruning, quantization

Languages & Activities

Languages: French (DALF C2); English and Arabic (fluent); Berber (native)
Interests: Robotics, Astronomy, Model Aircraft, Human–Machine Interfaces, Podcast Listening
Associative Life: Ex-IT Member at the IEEE Student Branch of the Polytechnic School of Algiers